2.7 Forgetting & other memory challenges

Why do we forget?

"if we rembered everything, we should on most occasions be as ill of as if we remembered everything."

Forgetting unimportant info helps us remember important info.

Forgetting bad memories is good for our mental well-being.

Some people have a kind of super memory highly superior autobiographical memory, where they remember incredibly accurately

this can be exhausting as one memory cues another

As we process info. we filter, alter or lose most

Sensory Memory → Working / Short-term memory → Long term Storage → Retrieval from LTs

- Sensory Memory: momentary register very many details
- Working / Short-term memory: few items are noticed & encoded
- Long term Storage: Some items altered/lost
- Retrieval from LTs: Some info gets retrieved, other info doesn't

Forgetting & the Two-track mind

Anterograde Amnesia → cant form new memories
Retrograde Amnesia → Cant recall old memories

Many with Rg. Amn. have a personality change but have stable sense of self
they think they have their old personalities

Anterograde Amnesia

Ppl w/ At. Amn. have no sense of elapsed time since getting Amn.

they can learn nonverbal tasks. Can quickly re-identify locations of hidden objects after finding it once or can remember path to bathroom

Can learn complicated procedural job skills

Can be clasically conditioned

But they do all this with no awareness of having learned them

Their automatic processing is intact

this Shows we have 2 distinct memory systems

Encoding Failure

Much of what we sense we fail to notice

What we fail to encode → we fail to remember

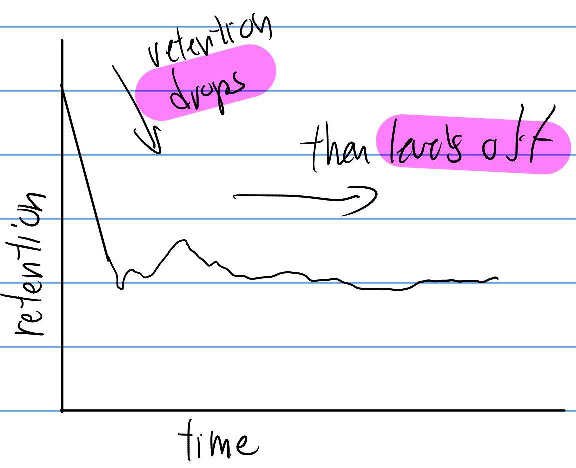
Stmem has limited capacity of few items at a time

Displacement ∴ info not encoded for LTS is lost
'pushed out' as new info enters STmem

Age can affect encoding efficiency

We selectively attend to very few of all of the many stimuli bombarding us → only a small fraction of stimuli has chance of being encoded

Storage Decay



Hermann Ebbinghaus came up with The Forgetting Curve.

Course of forgetting initially rapid then levels off

Study: 3 years out of school, most of what they learned is forgotten

But what they did remember, they likely would keep remembering 25+ years later

One explanation for this is the gradual fading of the physical memory trace

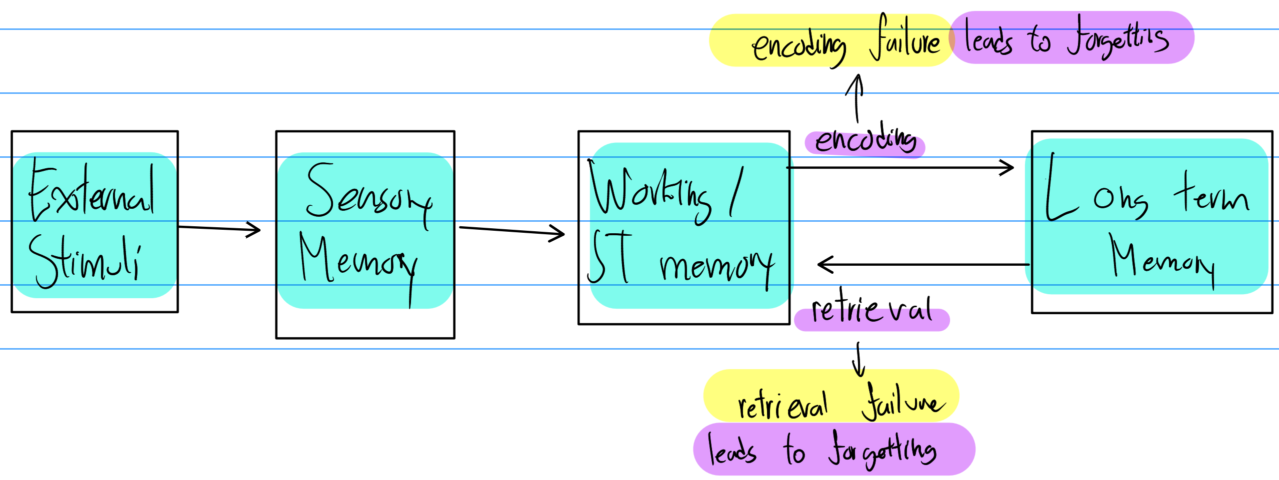
Retrieval Failure

In LT mem we store:

- important info
- rehearsed info

Memories unretrieved may be forgotten

Tip-of-the-tongue phenomenon → where we can't think of a word,



Interference

As you collect more info, your brain does not fill, but gets cluttered

Proactive Interference - Prior info disrupts new learning (forward-acting)

Retroactive Interference - new learning disrupts recall of old info (backward-acting)

Sleep

Info presented an hour before sleep, is better retained, suffering less retroactive interference

But seconds before sleep is seldom remembered

Basically nothing "learned" while asleep is remembered - but ears do register the audio

Positive transfer:

Old & new info dont always compete

Sometimes, previous info can facilitate learning of new info

Personal Connection

I frequently experience retroactive experience: I change passwords on important accounts often, leading to difficulty recalling which is the newest

Motivated Forgetting

To remember the past is to revise it

Our memory is an "unreliable, self-serving, historian"

Sigmund Freud thought we might repress painful & unforgettable memories

to protect our self-concept: ego & minimize anxiety

But the repressed memory would linger

He based his, very theoretical, theories on this

It's a popular idea, but most professionals think that it rarely, if even, occurs.

Trauma releases stress hormones, causing trauma survivors to attend to & remember the threat

So victims usually have intrusive, persistent memories they would like to, but can't, repress

Are our memories real?

No.

2/3 of americans think mem is like camera

mem actually very inaccurate

We often use constructive memory ∴ inferring our past from stored info
+ plus what we later imagined, expected, saw & heard

Mems are constructed: we don't retrieve, we just reweave

Mems are capable of constant revision

When we 'replay' a mem → we often replace original with slightly modified version

This process is called reconsolidation

'Your mem is only as good as your last mem. Fewer times used, more pristine it is'

∴ to a degree, all memory is false.

Misinformation & Imagination Effects

many eyewitnesses reconstruct mem after crime/accident

The wording of questions asked, ex. car 'hit' vs 'smashed' changed likelihood of recalling fictitious memories of broken glass

misinformation effect :: exposure to subtly misleading info → many confidently misremember what they saw/heard

~ 1/2 people vulnerable to misinfo effect

these false mems wither away once researchers debrief participants

misinfo effect can affect attitudes & behaviours

In experiment, researchers changed photos causing children to recall false info :: even a memorable hot-air balloon ride

Imagination inflation :: imagining an event can increase belief that event happened :: caused children to remember more vividly days later

Many ppl - 39% in one survey - recall improbable first mem from age ≤ 2

Source Amnesia

Source can be frailest part of memory

SrsAmn. is the faulty mem of how, when, or where info was learned or imagined

- we may cite stat, but can't recall source
- we may dream an event, later unsure if it was real
- may tell friend news, forget it was that friend who told us the news

Source Misattribution ∈ SrsAmn., where we misattribute the source of info to someone else, possibly ourselves

SrsAmn can explain déjà vu. 2/3 experienced this feeling
the eerie sense 'I've been in this situation before'

Cause may be familiarity with a stimulus + uncertainty of where its from

Normally, first, we exp. familiarity (temp. lobe processing), and then consciously remember details (Hippocampus + front. lobe processing)

When these functions & regions are out of sync, we exp. familiarity w/o conscious recall : brain does its best to make sense of odd moment

Discerning True & False memories

unreal memories feel real.

misinfo effect + SrcAmn happen outside conscious awareness
∴ hard to seperate true vs. false

more retelling of a memory → more filling in gaps with reasonable guesses & assumptions → those guessed details feel as real as rest of mem

We more easily remember gist than exact words

False mems are contagious

We forget where we learned info

This is how false info spreads so easily

This is how people get unfairly sent to prison or, a recent ex., be murdered by the state, for crime not committed

Other-race effect → better recall faces of our race than other races

40% of exonerated were falsely accused after cross-racial misidentification

Mem construction errors can explain why "'hypnotically' refreshed' mems of crimes so easily incorporate falsehoods

people in relationships overestimate first impression of partner after breaking up → underestimate first impressions of ex

This may be a reason why many people think they were not fascists some time ago, as their mem has been influenced by current re-educated views

"Maturation makes liars of us all"

How reliable are children eyewitnesses?

Problem: Child abuse is extremely serious issue
Children's memories very easily molded

in a study: 58% of preschoolers produced false but often extremely vivid memories they've never exp.

prof. psycho. could not reliably seperate the false mems from real ones nor the children themselves

Children regularily identified innocent suspects as guilty

Solution: with carefully trained interviewers, adults & children can be accurate eye witnesses

To improve accuracy, esp. in children:

- use neutral words they understand; ask nonleading questions
less suggestive; more effective

- get children not to talk w/ involved adults prior to interview

- disclosure was made in a first interview with a neutral person

Personal Connection

I am a frequent ChatGPT user as it is a powerful learning tool. I've learned similar concepts, esp. asking non-leading questions & neutral words, are best at keeping it from hallucinating, and especially for complex questions using divergent thinking

How can this be used practically?

Rehearse Repeatedly

Spacing effect - use distributed (spaced) practice

Engage in many seperate study sessions

Take advantage of little intervals to practice w/ Anki! not Quizlet 

Experts recommend retrieving item 3 times before you stop studying

testing effect - study actively

mentally saying, writing or typing > silently reading

production effect: we often learn something best when teaching it, when explaining it to ouselves, or when rehearsing out loud.

Personal Connection

There is a famous idea among computer programmers, making use of the production effect: the coding duck.
Simply the idea is that you explain your code to the duck to better understand it yourself.

Make Material Meaningful

Personalize the material

Build network of retrieval cues by forming as many associations as u can

Apply Concepts to your own life (literally this submodule)

Understand and organize info

relate material to other known concepts

try drawing the concept

Activate Retrieval Cues

recall importance of: context-dependent memory & state dependent memory

mentally recreate situation & mood where original learning occured

Use mnemonic devices

Make up a story w/ vivid imagery of the concepts

Chunk information & Create memorable mnemonic

Minimize proactive & retro active interference

don't schedule back-to-back studying for topics likely to interfere w/ each other

Sleep more

In sleep, brain reorganizes & consolidates info for long-term memory

Sleep helps u remember what u learned + what ur planning to do

Test ur own knowledge

both to
- rehearse it
- find out what you don't know

stops over confidence from ability to recognize

test recall

MCQ:

1. c 2. d 3. c
4. d 5. a 6. a
?